

# The Airfryer Market: A Data-Driven Market Intelligence Analysis

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# What We Studied

- Amazon purchase data, 2019–2023
- 10 leading brands: Chefman, Cosori, Cuisinart, Dash, GoWise USA, Instant Pot, Ninja, NuWave, Oster, Ultrean
- 50 brand-year observations

The air fryer category saw rapid adoption and is now maturing. Understanding who leads on price, ratings, and share, and why, reveals where value is being created and competed away.

# Prices, Ratings & Shares Over Time

## Who's Expensive? Who's Big?

Pricing (avg. price across 2019–2023)

Premium tier: Cuisinart (~\$229), Oster, Ninja, NuWave

Value tier: Dash (~\$55), Ultrean (~\$73)

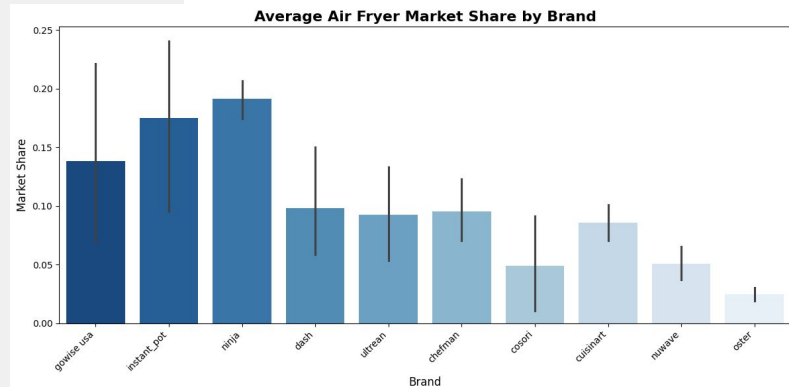
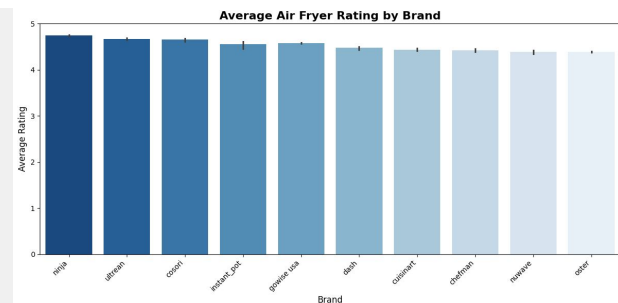
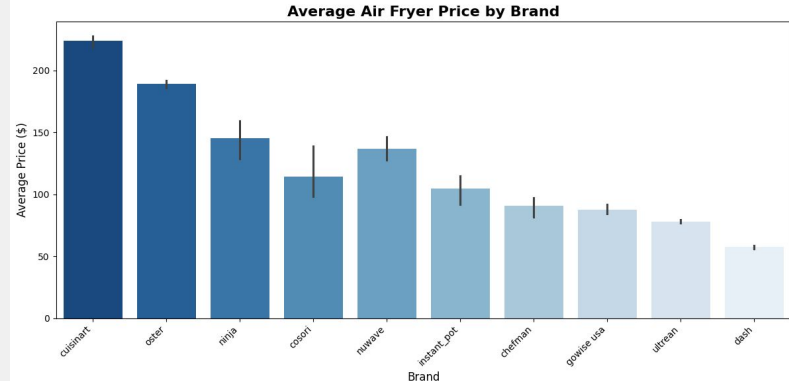
## Ratings (avg. customer review)

Ratings are tightly clustered across brands (~4.3–4.6)

No brand dramatically outperforms on satisfaction

## Market Share

GoWise USA, Ninja, and Instant Pot hold the largest shares



# Demand Model

## How we estimated price sensitivity:

- **Year dummies**  
absorb market-wide shifts in total demand
- **Categories**  
absorb stable brand-level preferences
- One shared **price coefficient** across all brands and years

## The Model (Multinomial Logit, estimated via OLS)

$$\log(s_{bt}) = \alpha_0 + \alpha_t + \gamma_b + \beta_{\text{price}} \cdot p_{bt} + \beta_{\text{rating}} \cdot r_{bt} + \sum_{\ell} \beta_{\ell} x_{bt\ell} + \varepsilon_{bt}$$

+

higher prices increase market share

—

higher prices reduce market share

## Key result: Price coefficient is NEGATIVE

- A higher price reduces a brand's market share, as economic theory predicts
- A positive coefficient would indicate model failure and make cost inference impossible

# The Price Coefficient

The price coefficient is  $\beta_{\text{price}} = -0.0377$ , meaning price has a negative relationship with share. A negative coefficient matters because this model correctly portrays how demand should behave. According to economic theory, when prices rise, demand falls.

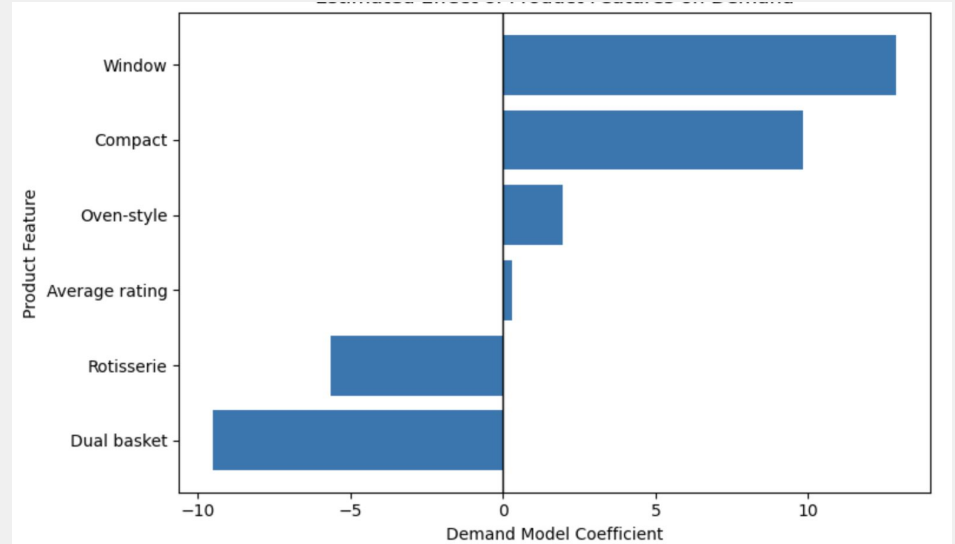
Price sensitivity is negative, which makes the model economically believable

# Most Important Product Features

## Key Findings:

- ★ Window-style air fryers seem to have the strongest positive association with demand
- ★ Compact models are also strongly associated with higher demand
- ★ Rotisserie and dual-basket features are negatively associated with demand

Overall, customers seem to respond better to practical and more visible design features.



# Unit Costs, Markups, and Share-Weighted Profit

01

|             | unit_cost | markup |
|-------------|-----------|--------|
| brand       |           |        |
| chefman     | 61.57     | 29.37  |
| cosori      | 86.28     | 27.98  |
| cuisinart   | 194.90    | 29.05  |
| dash        | 27.94     | 29.54  |
| gowise usa  | 56.30     | 31.15  |
| instant_pot | 71.90     | 32.56  |
| ninja       | 112.49    | 32.85  |
| nuwave      | 109.05    | 27.97  |
| oster       | 161.80    | 27.23  |
| ultrean     | 48.79     | 29.33  |

Cuisinart and Oster have the highest estimated unit costs while Dash and Ultrean operate at the lowest cost levels.

02

## Markups

Markups appear similar across brands (around \$27 to \$33). However, Ninja and Instant Pot seem to have the highest average markups. This is likely due to competition between brands reducing the limits on how much brands are able to charge for extra profit.

03

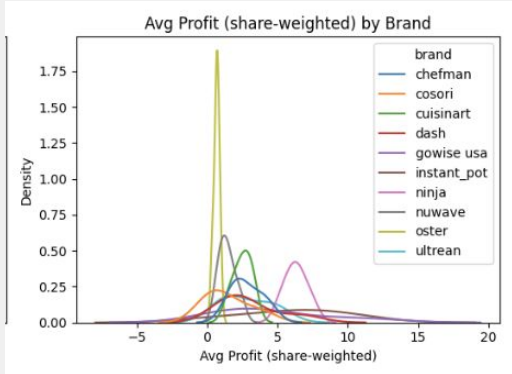
## Share-Weighted Profit

|             |        |
|-------------|--------|
| brand       |        |
| ninja       | 6.3061 |
| instant_pot | 6.0103 |
| gowise usa  | 4.6058 |
| dash        | 2.9944 |
| chefman     | 2.8210 |
| ultrean     | 2.7809 |
| cuisinart   | 2.5030 |
| cosori      | 1.4367 |
| nuwave      | 1.4229 |
| oster       | 0.6783 |

Combining markup and brand market share, Ninja and Instant Pot excel through meaningful market presence and reasonable markups.



# Conclusion: Brand Positioning



**Share-weighted profit showing  
the strongest brands combine  
demand share with healthy  
markups**

## Overall Market Leaders



- Ninja and Instant Pot are the strongest overall brands due to their high share-weighted profit
- They are able to effectively balance a strong market demand with healthy markups to generate profits despite competition

## Market Share Leaders



- GoWise has one of the largest market shares with profitability stemming from volume sold and pricing strategy
- Appears less premium than Ninja and Instant Pot

## Value Players



- Dash, Ultra, and Chefman compete for the most affordability as lower estimated costs help them stay competitive in the fragmented market

|   | category   | year | brand      | purchase_count | product_count | avg_price  |
|---|------------|------|------------|----------------|---------------|------------|
| 0 | air_fryers | 2019 | chefman    | 1146           | 10            | 72.963695  |
| 1 | air_fryers | 2019 | cosori     | 11             | 2             | 159.990000 |
| 2 | air_fryers | 2019 | cuisinart  | 1616           | 22            | 229.465274 |
| 3 | air_fryers | 2019 | dash       | 3011           | 19            | 55.176333  |
| 4 | air_fryers | 2019 | gowise usa | 4405           | 45            | 83.575551  |

## Premium



- Cuisinart, Oster, and NuWave are closer to the premium part of the market but do not achieve the same profit position as stronger premium players



Thank you!